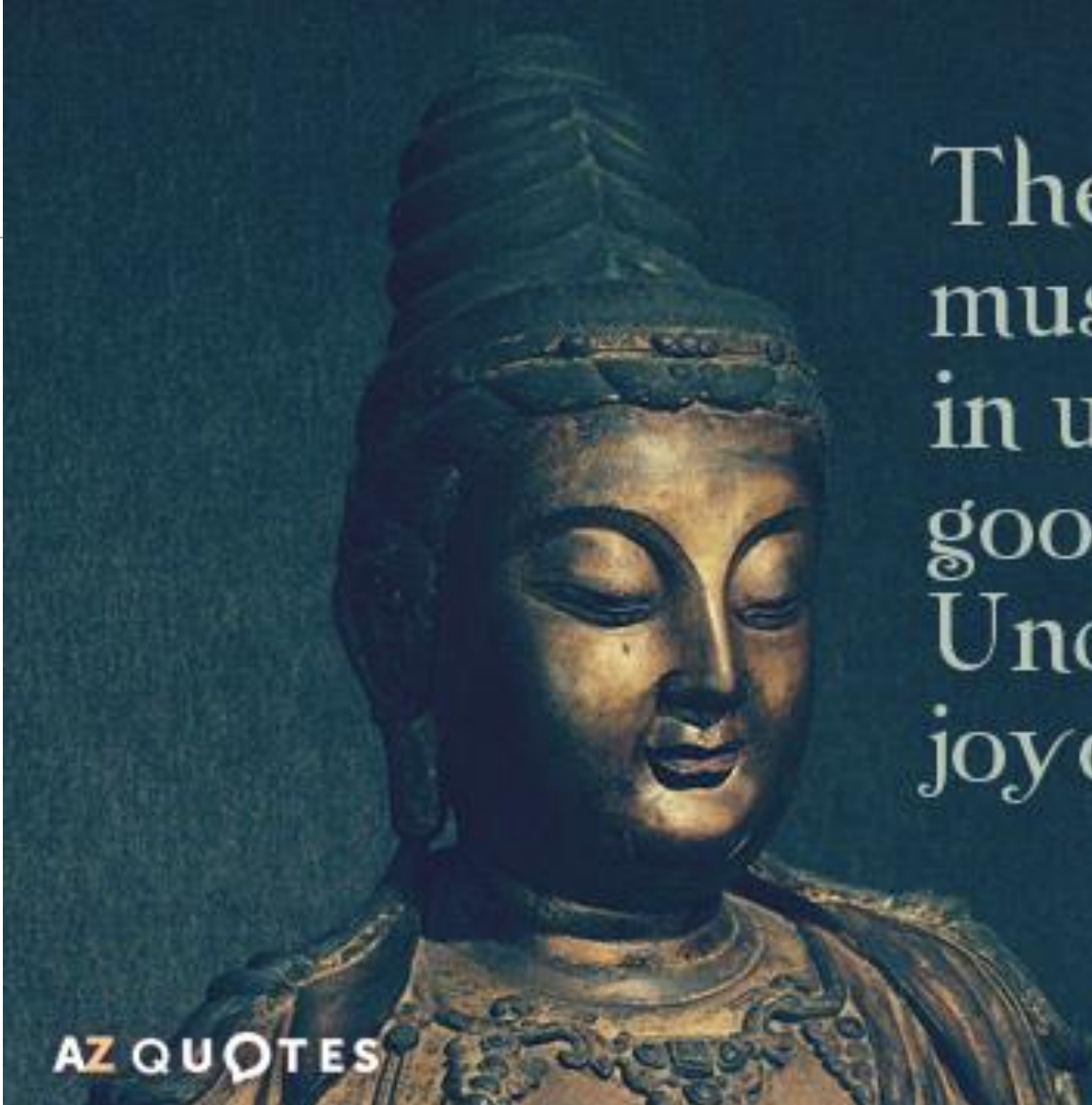




The brain is like a
muscle. When it is
in use we feel very
good.
Understanding is
joyous.

Carl Sagan

AZ QUOTES

Deep Learning

Dr. Pravin Rahate

Department of Computer Engineering

Fr. C.Rodrigues Institute of Technology, Vashi

Neural Networks are taking over!

- Neural networks have become one of the main approaches to AI
- They have been successfully applied to various pattern recognition, prediction, and analysis problems
- In many problems they have established the state of the art

Breakthroughs with neural networks

www.technewsworld.com/story/54013.html

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Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

Print Email



Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

How do you feel about Black Friday and Cyber Monday?

- ☐ They're great -- I get a lot of bargains!
- ☐ The deals are too spread out -- I'd prefer just one day.
- ☐ They're a fun way to kick off the holiday season.
- ☐ I don't like the commercialization of Thanksgiving Day.
- ☐ They're crucial for the retail industry and the economy.
- ☐ The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

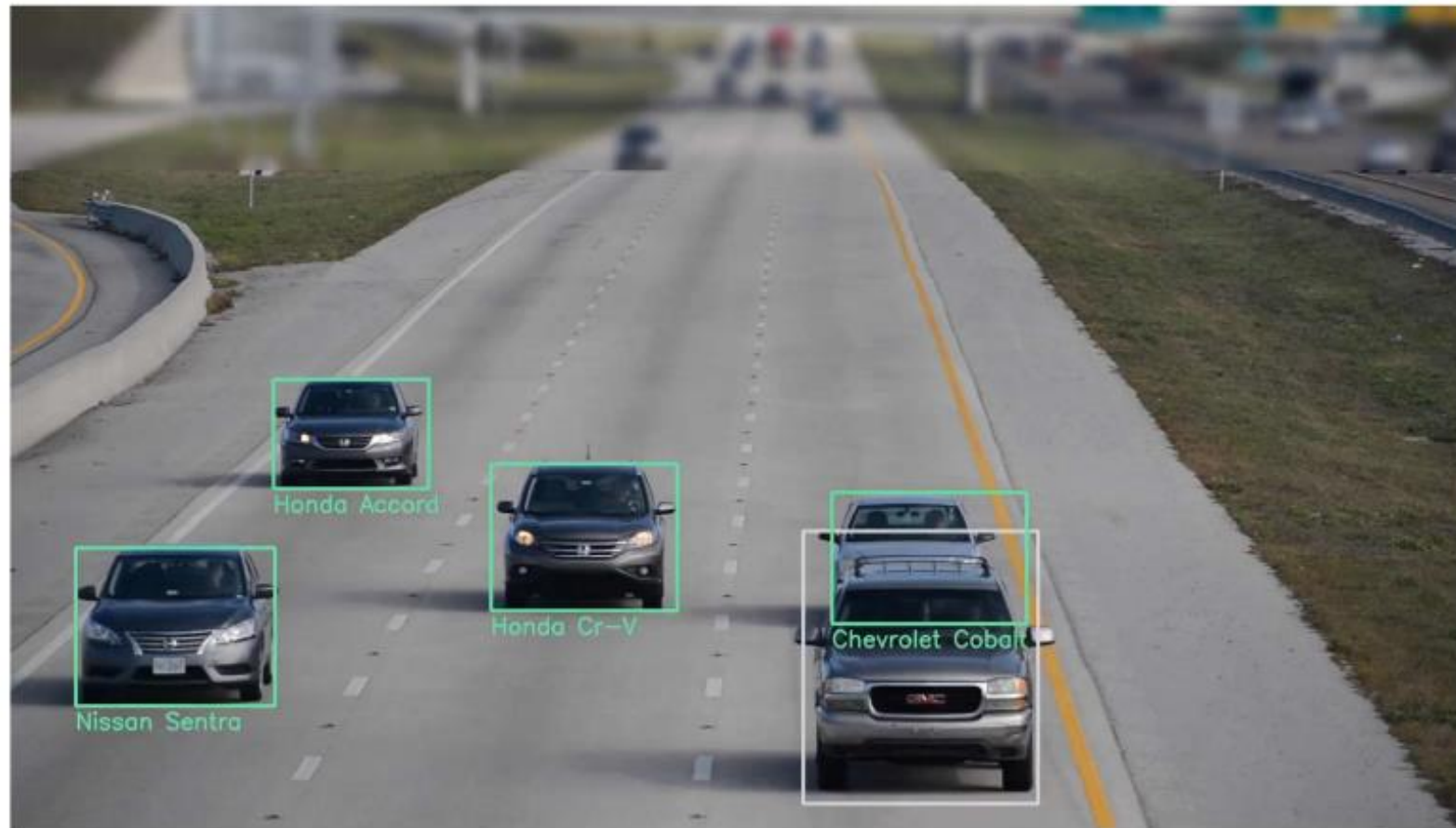
AWS Enjoys Top Perch in IaaS, PaaS Markets

US Comptroller Gears Up for Blockchain and

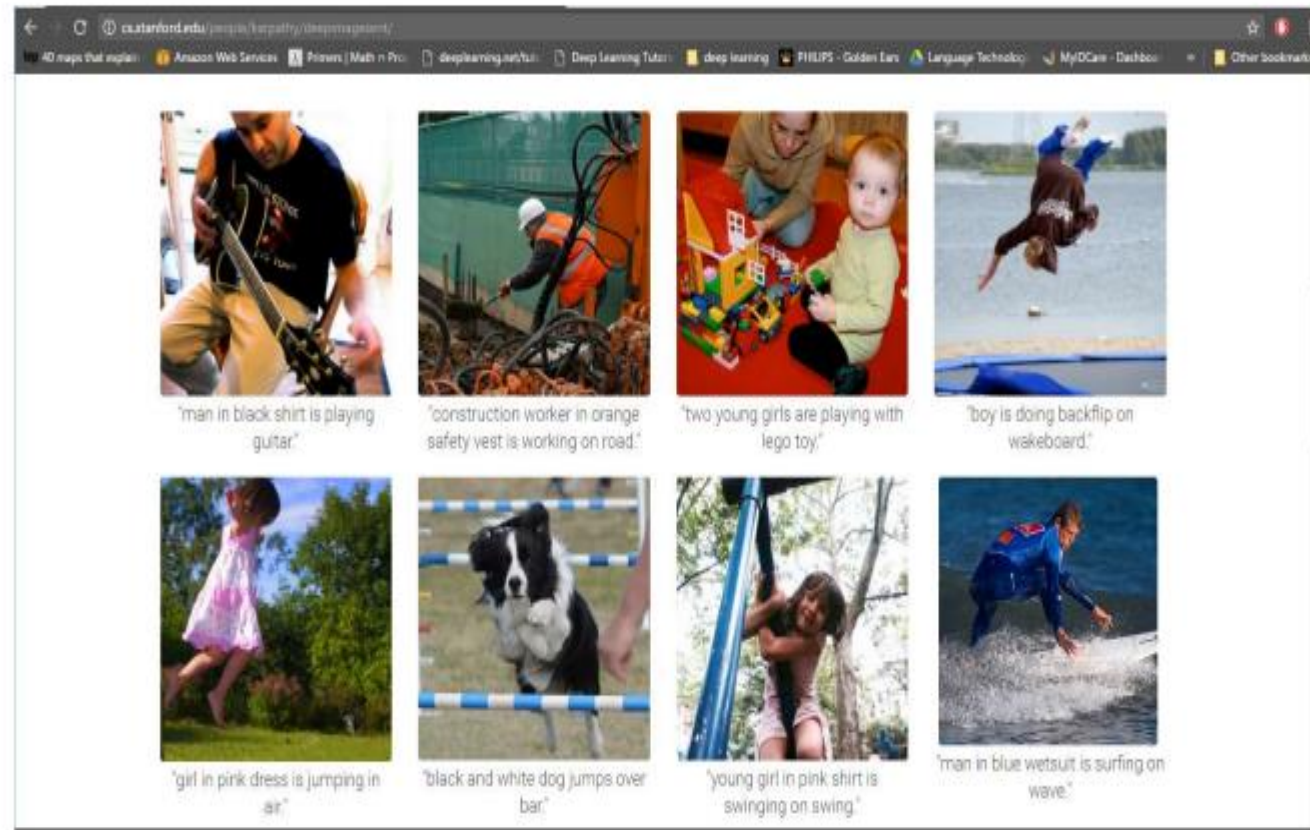
Breakthroughs with neural networks



Image Recognition



Success with neural networks



Success with neural networks

And a variety of other problems:

- From art to astronomy to healthcare..
- and even predicting stock markets!

Neural nets and the employment market



This guy didn't know
about neural networks
(a.k.a deep learning)



This guy learned
about neural networks
(a.k.a deep learning)

Fundamentals of Neural Network

It begins with this..

The Brain

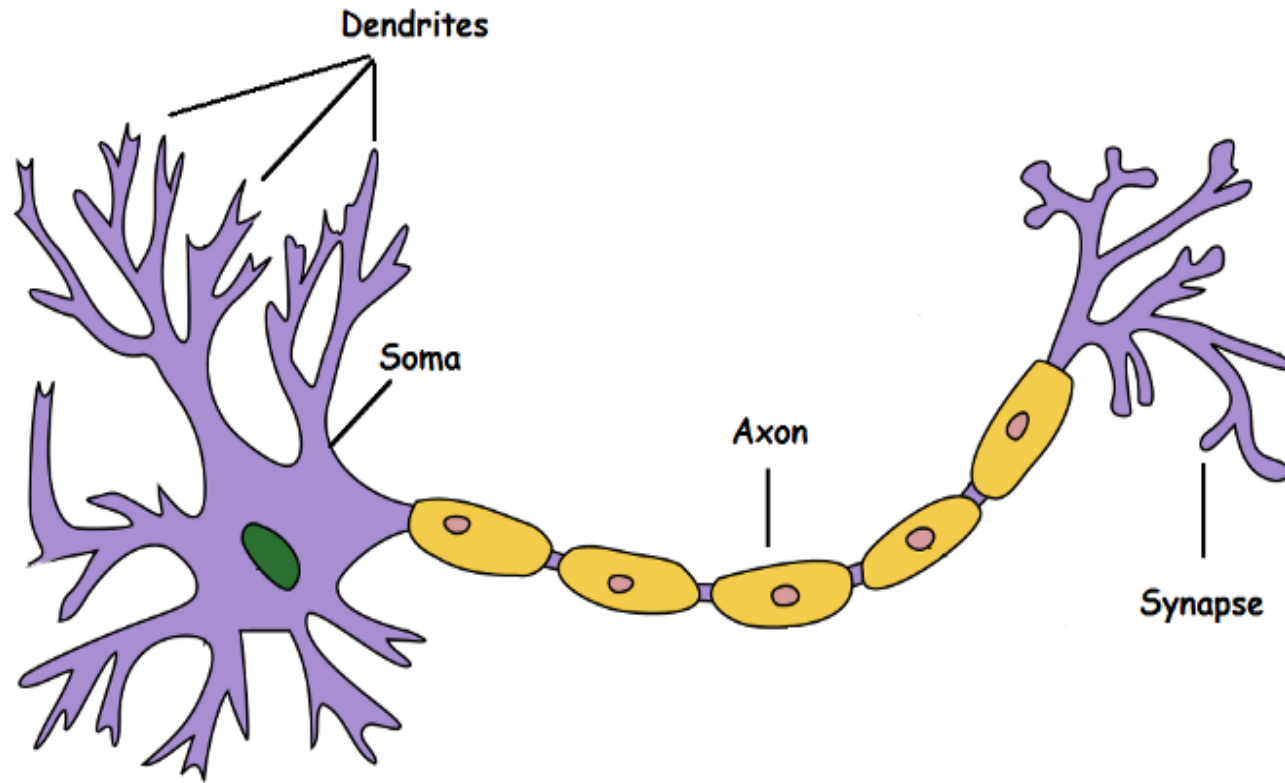


Brain: Interconnected Neurons



- Many neurons connect *in* to each neuron
- Each neuron connects *out* to many neurons

Biological Neurons



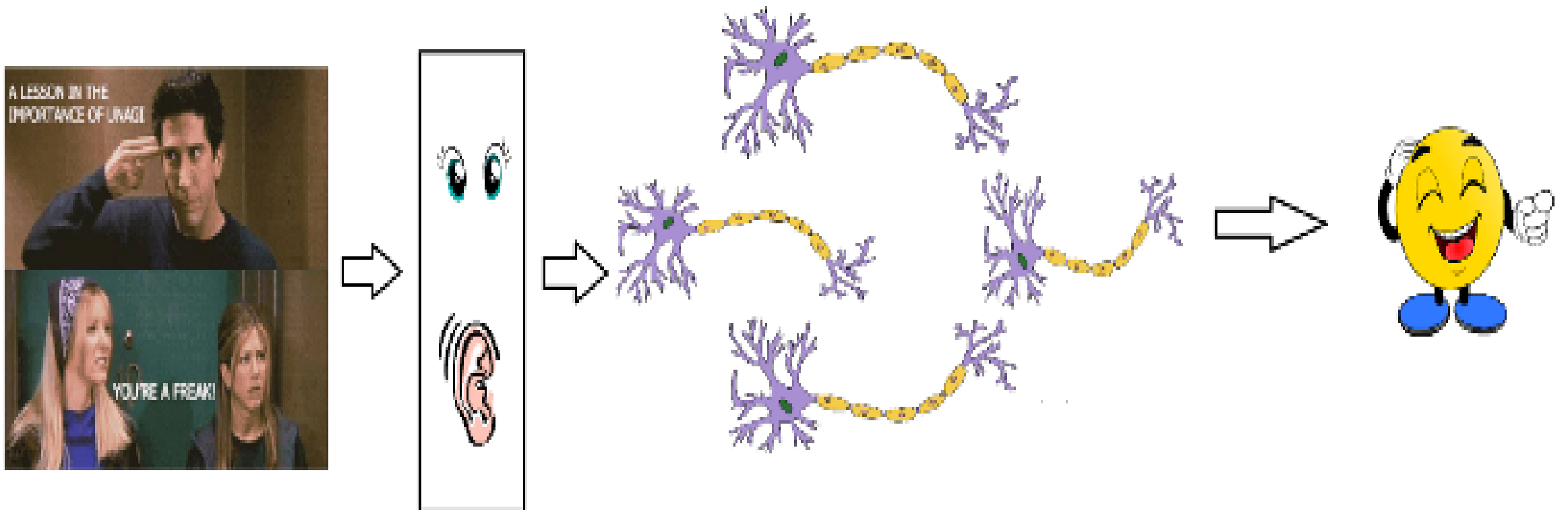
Dendrite: Receives signals from other neurons

Soma: Processes the information

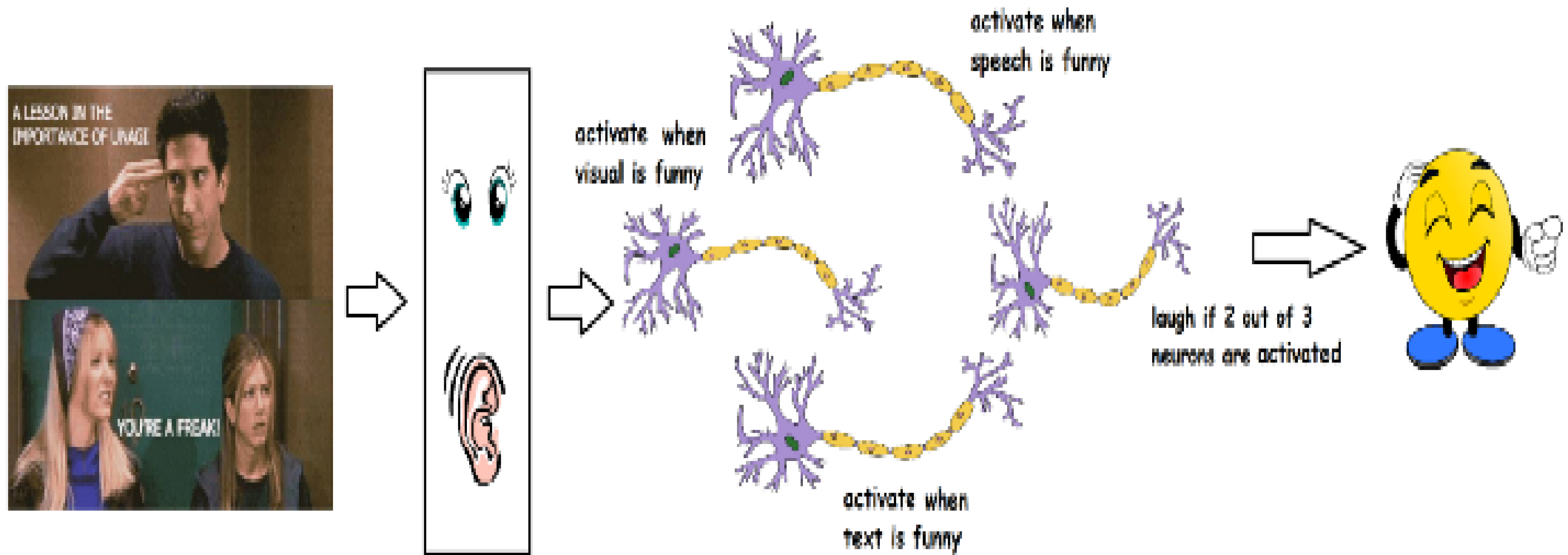
Axon: Transmits the output of this neuron

Synapse: Point of connection to other neurons

Biological Neurons

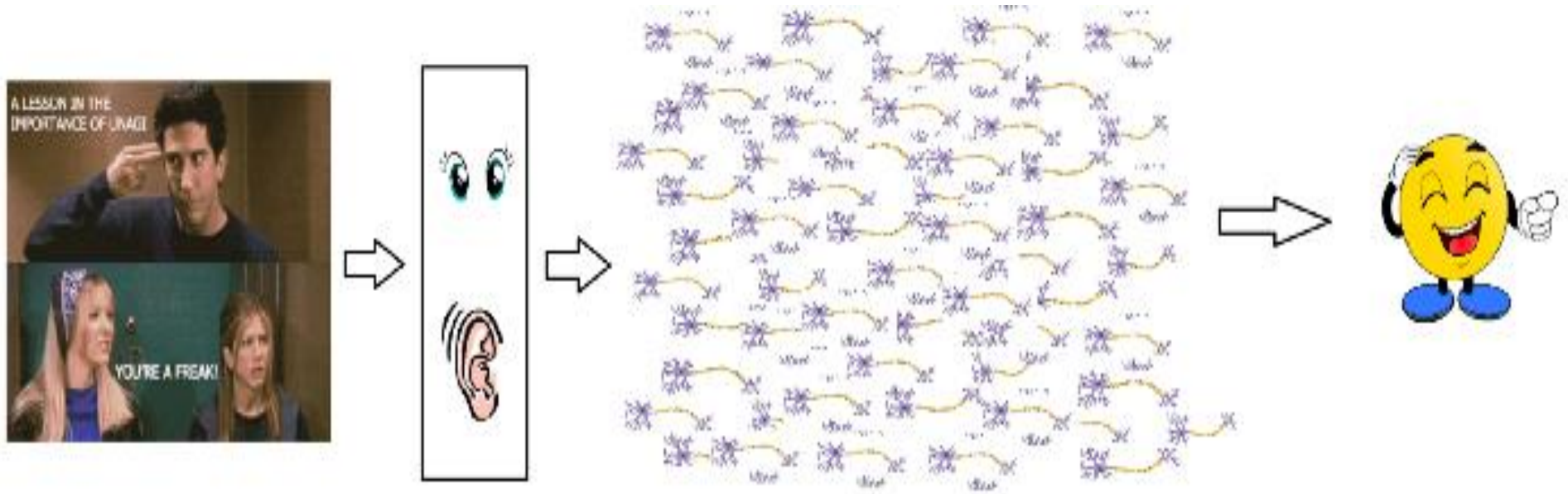


Biological Neurons – Division of work



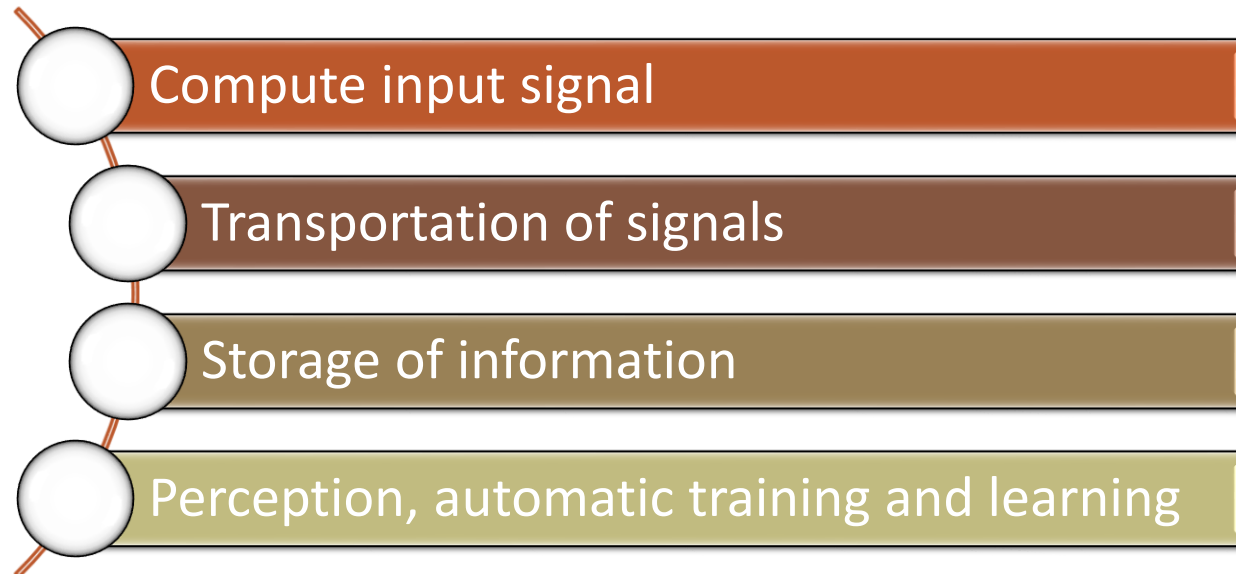
Biological Neurons – Decision Making

- In reality, it is not just a couple of neurons which would do the decision making.
- There is a massively parallel interconnected network of 10¹¹ neurons (100 billion) in our brain

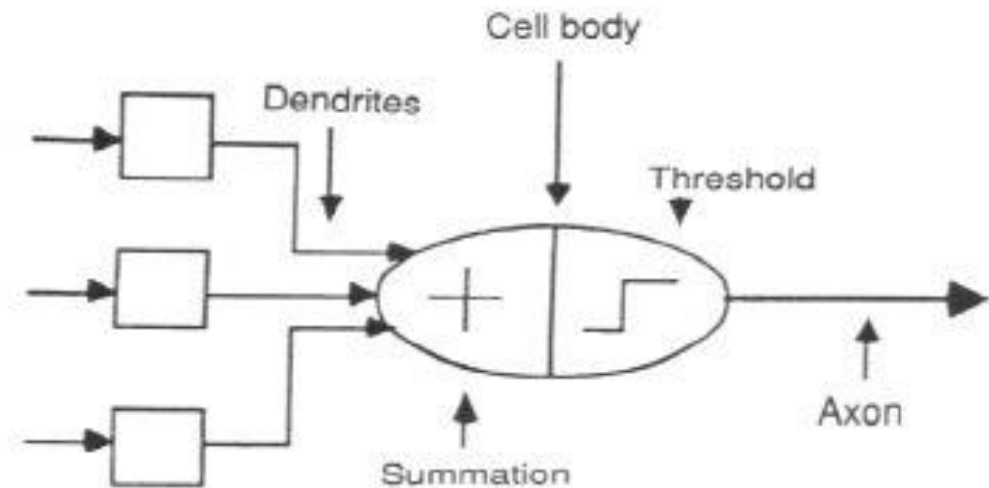
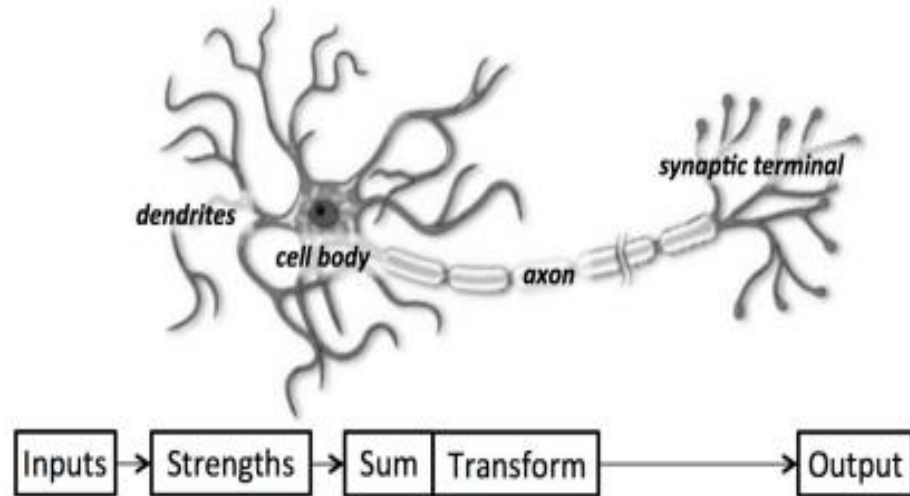


Artificial Neural Network (ANN)

- ❑ Artificial neural networks (ANNs) or neural network (NNs), are simplified models (i.e. imitations) of the biological nervous system.
- ❑ We know neuron is the integral part of interconnected network of nervous system.
- ❑ It serves the following –

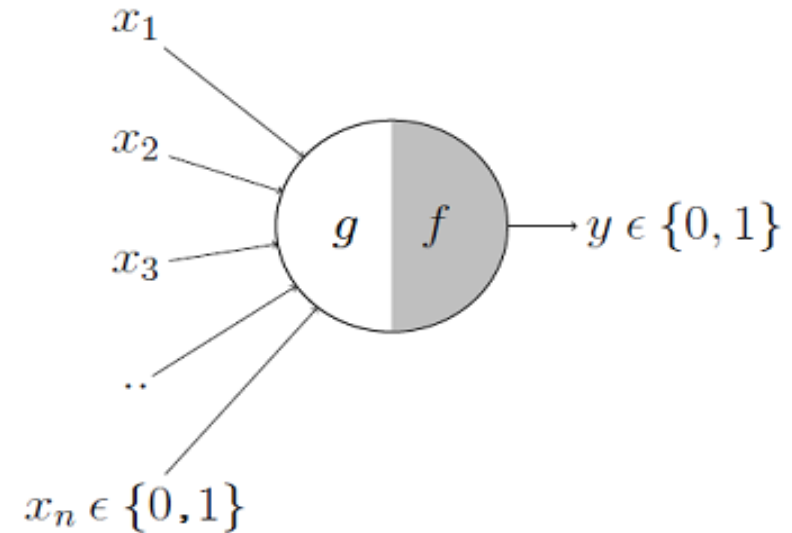


Association of BNN with ANN



McCulloch-Pitts Neuron

- This is where it all began..
- The first computational model of a neuron was proposed by Warren McCulloch (neuroscientist) and Walter Pitts (logician) in 1943
- It may be divided into 2 parts.
- The first part, g takes an input (ahem dendrite ahem) & performs an aggregation
- Based on the aggregated value the second part, f makes a decision.



Example

- I want to predict my own decision, whether to watch a random football game or not on TV.
- The inputs are all boolean i.e., {0,1}
- my output variable is also boolean {0: Will watch it, 1: Won't watch it}.

x_1 could be *isPremierLeagueOn* (I like Premier League more)

x_2 could be *isItAFriendlyGame* (I tend to care less about the friendlies)

x_3 could be *isNotHome* (Can't watch it when I'm running errands. Can I?)

x_4 could be *isManUnitedPlaying* (I am a big Man United fan. GGMU!) and so on.

Example

- These inputs can either be excitatory or inhibitory.
- Inhibitory inputs are those that have maximum effect on the decision making irrespective of other inputs
i.e., if x_3 is 1 (not home) then my output will always be 0 i.e., the neuron will never fire, so x_3 is an inhibitory input.
- Excitatory inputs are NOT the ones that will make the neuron fire on their own but they might fire it when combined together.

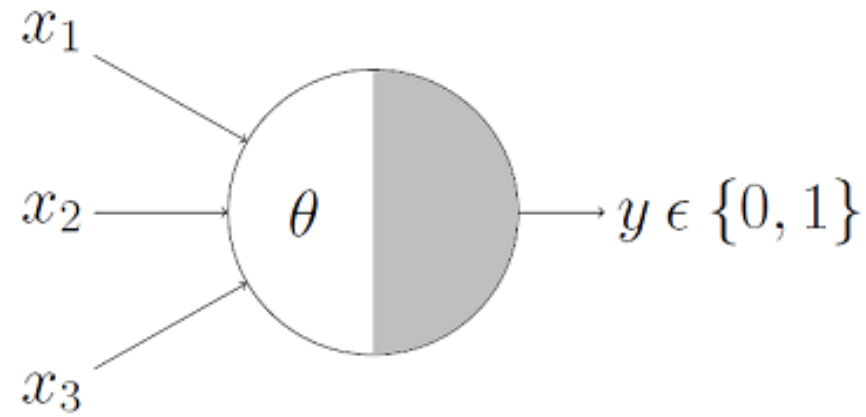
Example

$$g(x_1, x_2, x_3, \dots, x_n) = g(\mathbf{x}) = \sum_{i=1}^n x_i$$

$$\begin{aligned} y = f(g(\mathbf{x})) &= 1 \quad \text{if } g(\mathbf{x}) \geq \theta \\ &= 0 \quad \text{if } g(\mathbf{x}) < \theta \end{aligned}$$

- **$g(\mathbf{x})$** is just doing a sum of the inputs — a simple aggregation.
- ***theta*** here is called thresholding parameter.
 - For example, if I always watch the game when the sum turns out to be 2 or more, the ***theta*** is 2 here.

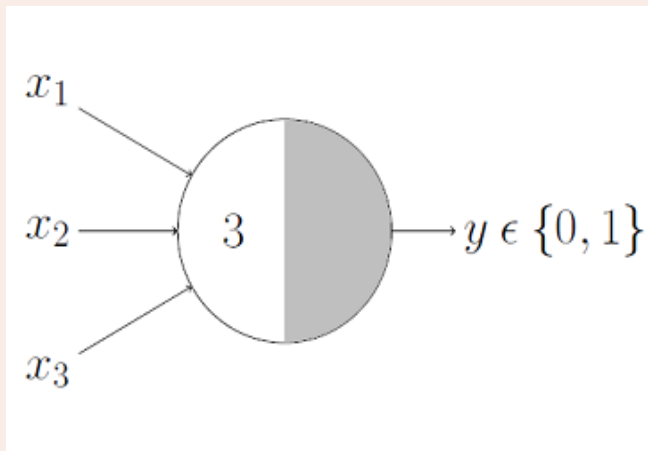
M-P Neuron: A Concise Representation



- This representation just denotes that, for the boolean inputs x_1 , x_2 and x_3 if the $g(x)$ i.e., $\text{sum} \geq \text{theta}$, the neuron will fire otherwise, it won't.

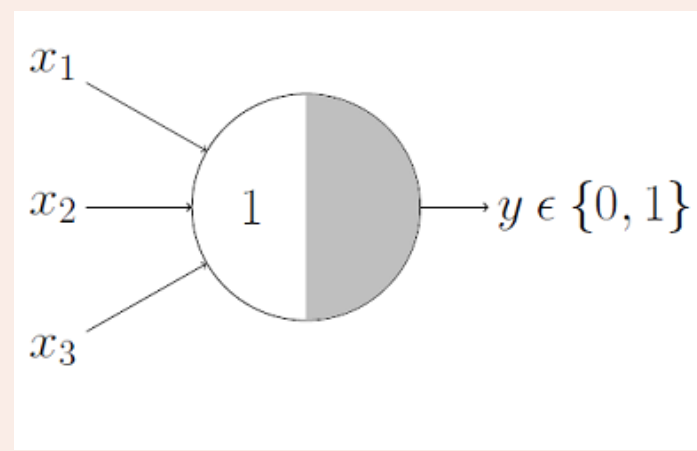
M-P Neuron: Logic gate

AND Function



An AND function neuron would only fire when ALL the inputs are ON i.e., $g(x) \geq 3$ here.

OR Function



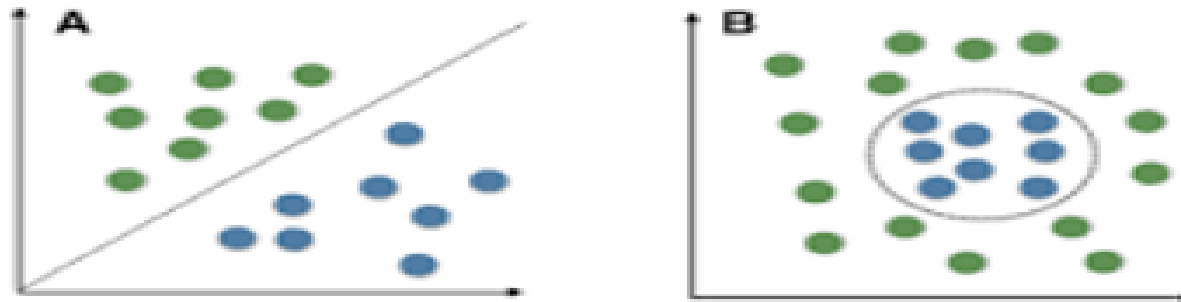
OR function neuron would fire if ANY of the inputs is ON i.e., $g(x) \geq 1$ here.

Limitations Of M-P Neuron

- ❑ What about non-Boolean (say, real) inputs?
- ❑ Do we always need to hand code the threshold?
- ❑ Are all inputs equal? What if we want to assign more importance to some inputs?
- ❑ What about functions which are not linearly separable? Say XOR function.

Linearly Separable & Non Linearly Separable Classes

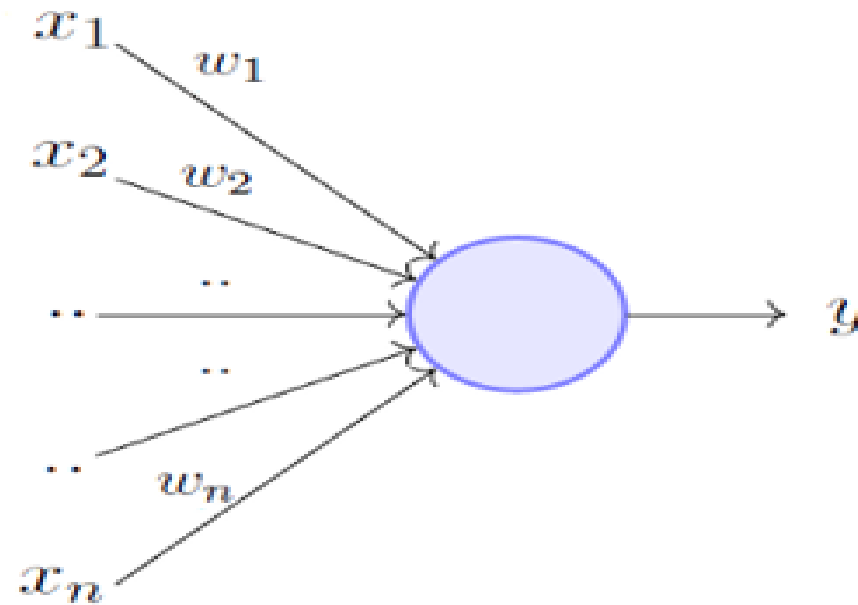
- ❑ Linear Classification refers to categorizing a set of data points into a discrete class based on a linear combination of its explanatory variables.
- ❑ Non-Linear Classification refers to categorizing those instances that are not linearly separable



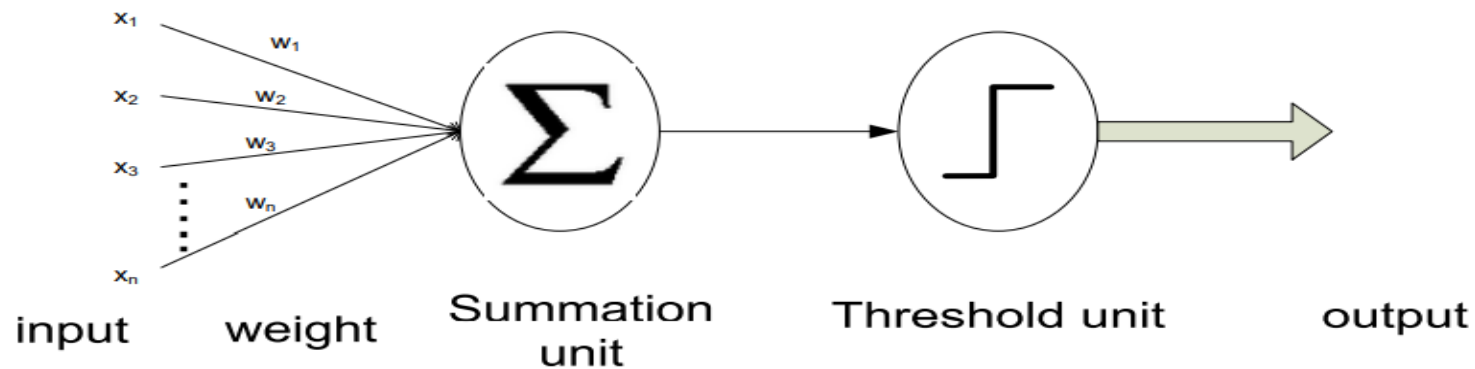
Perceptron Model

- Overcoming the limitations of the M-P neuron, Frank Rosenblatt, an American psychologist, proposed the classical perception model, the mighty artificial neuron, in 1958.
- Further refined and carefully analyzed by Minsky and Papert (1969) — their model is referred to as the perceptron model.

Perceptron Model



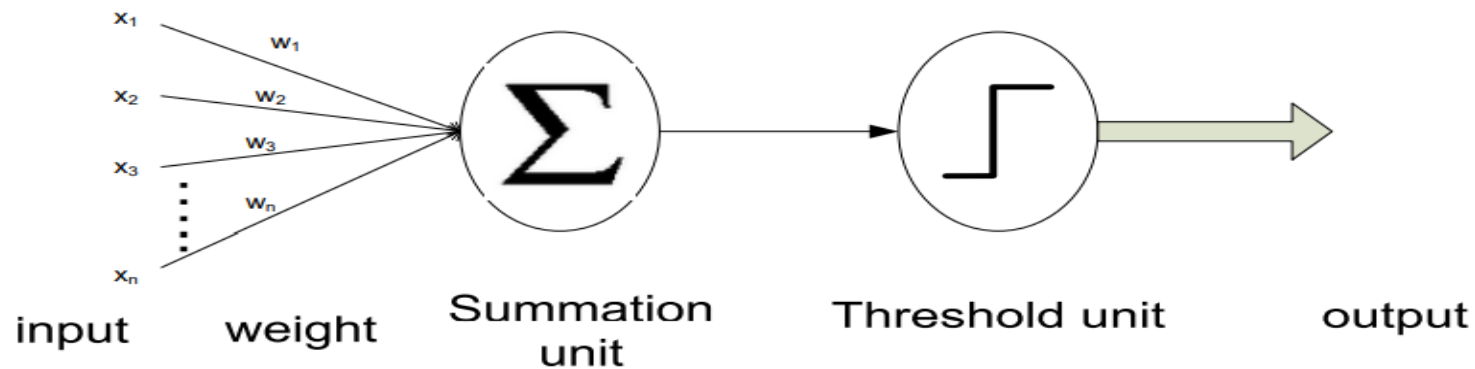
Perceptron Model – Step 1 – Setting weights



Here, $x_1, x_2, \dots, x_n$ are the n inputs to the artificial neuron.

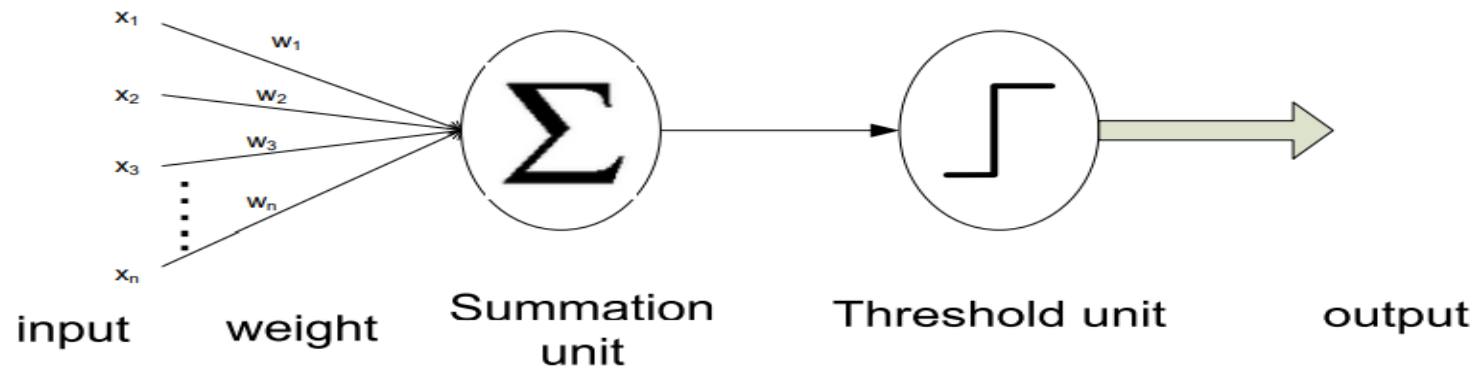
$w_1, w_2, \dots, w_n$ are weights attached to the input links.

Perceptron Model- Step 2 - Summation



$$w_1 x_1 + w_2 x_2 + \dots + w_n x_n = \sum_{i=1}^n w_i x_i$$

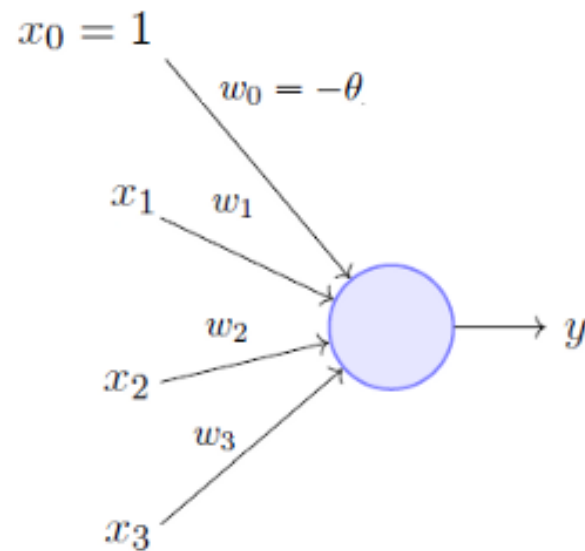
Perceptron Model – Step 3 - Activation



To generate the final output y , the sum is passed to a **filter f** called [transfer function](#), which releases the output.

$$y(x) = f\left(\sum_{i=1}^n w_i x_i\right)$$

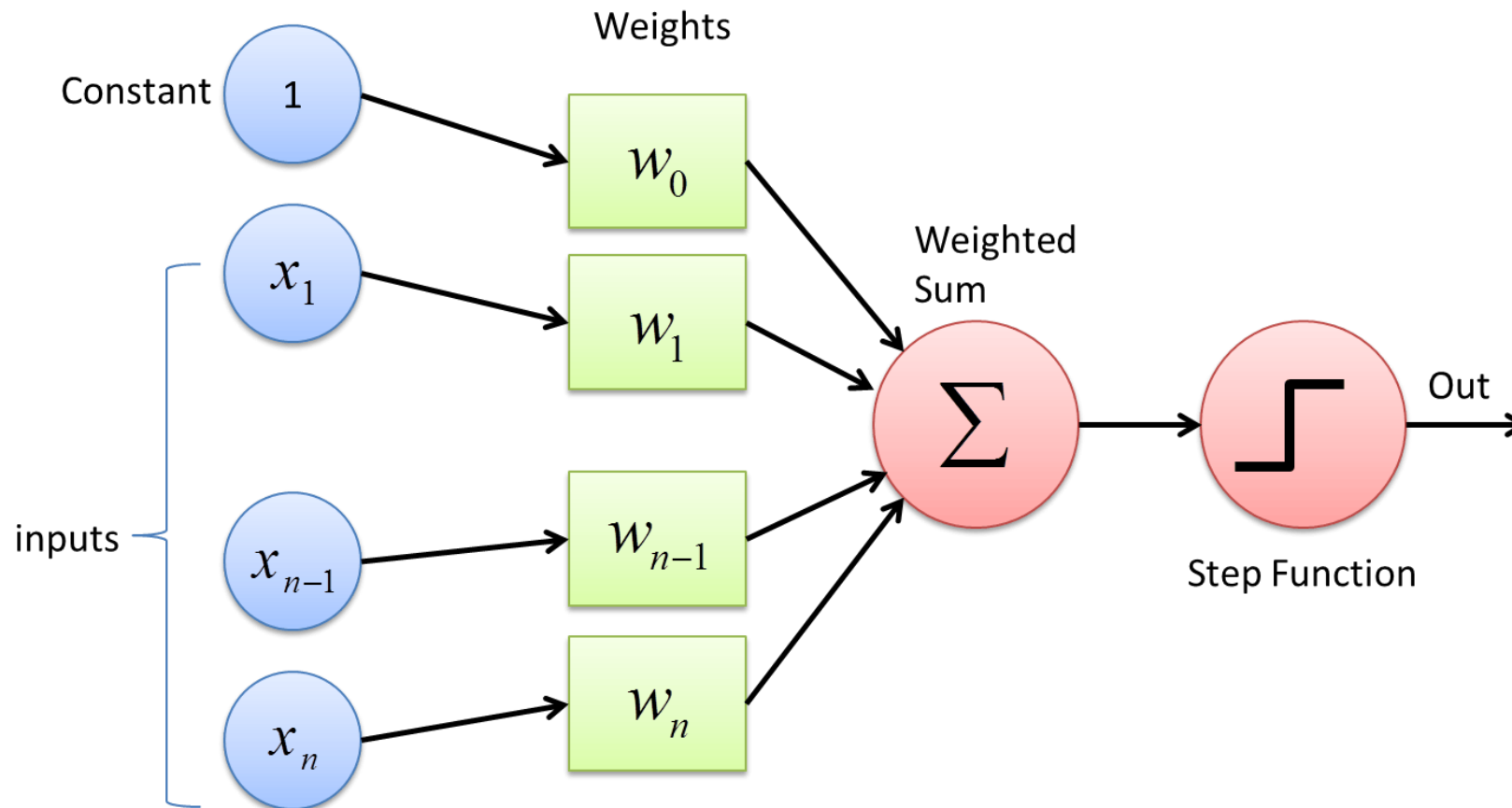
Perceptron Model – Example



$x_1 = isPremierLeagueOn$
 $x_2 = isManUnitedPlaying$
 $x_3 = isFriendlyGame$

Here, w_0 is called the **bias** because it represents the prior (prejudice).

Perceptron Model – Example



Perceptron Model – OR function

OR Function — Can Do!

x_1	x_2	OR	
0	0	0	$w_0 + \sum_{i=1}^2 w_i x_i < 0$
1	0	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
0	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 0 < 0 \implies w_0 < 0$$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 1 \geq 0 \implies w_2 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 0 \geq 0 \implies w_1 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 1 \geq 0 \implies w_1 + w_2 > -w_0$$

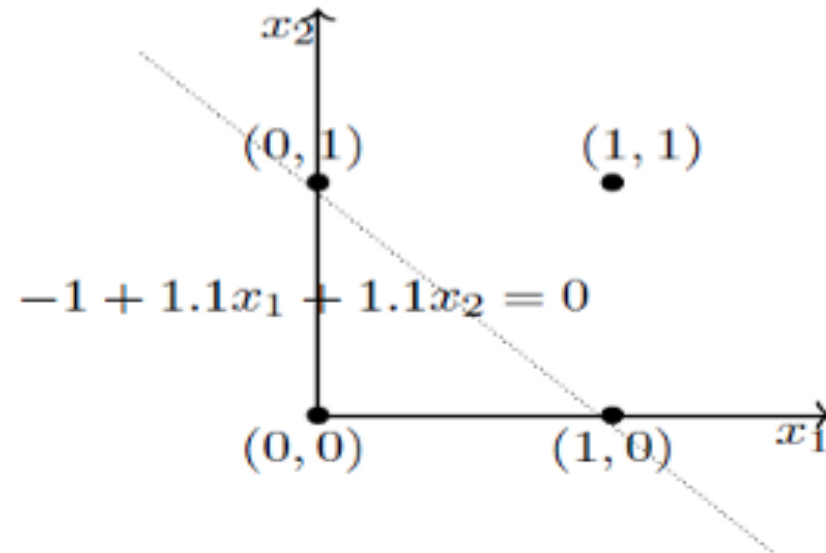
Perceptron Model – OR function

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0	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$

One possible solution is

$$w_0 = -1, w_1 = 1.1, w_2 = 1.1$$



Perceptron vs McCulloch-Pitts Neuron

McCulloch Pitts Neuron
(assuming no inhibitory inputs)

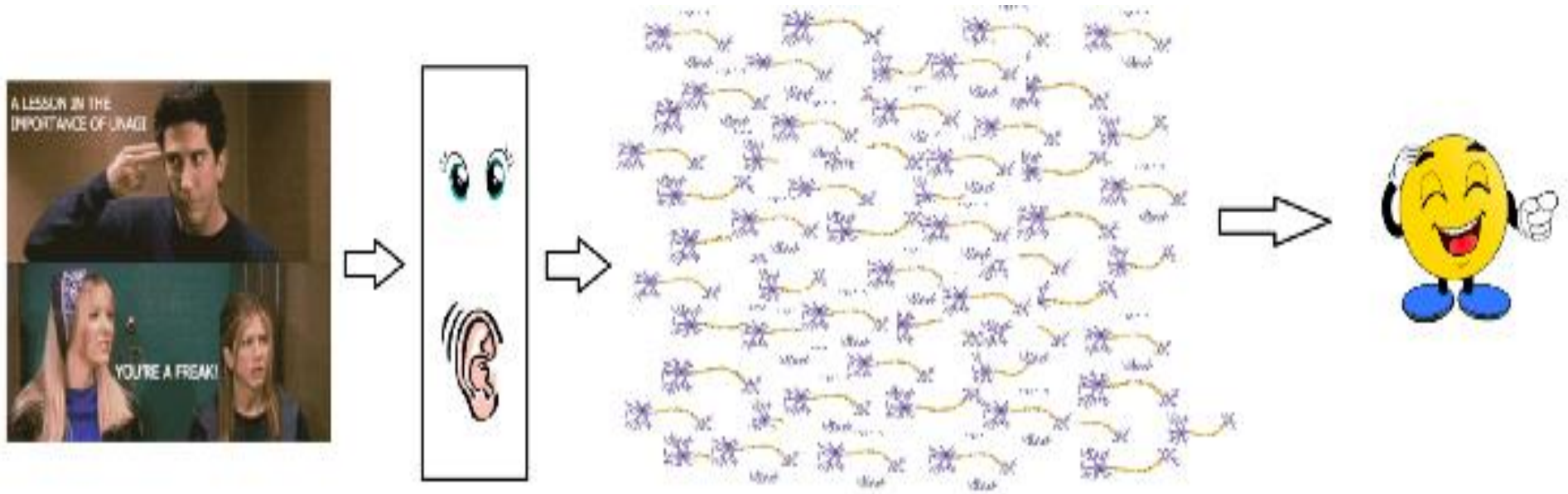
$$y = 1 \quad \text{if } \sum_{i=0}^n x_i \geq 0$$
$$= 0 \quad \text{if } \sum_{i=0}^n x_i < 0$$

Perceptron

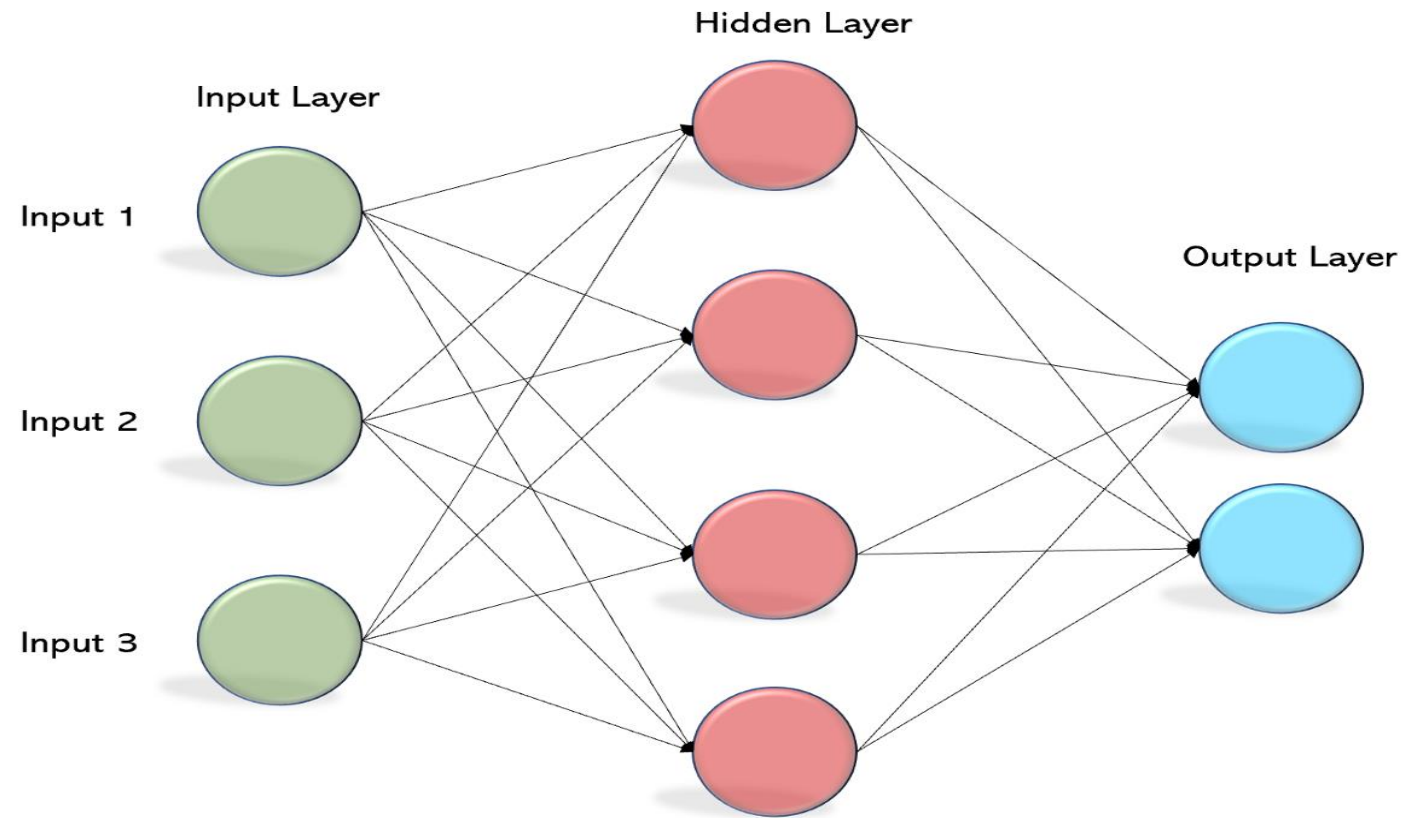
$$y = 1 \quad \text{if } \sum_{i=0}^n w_i * x_i \geq 0$$
$$= 0 \quad \text{if } \sum_{i=0}^n w_i * x_i < 0$$

Biological Neurons – Decision Making

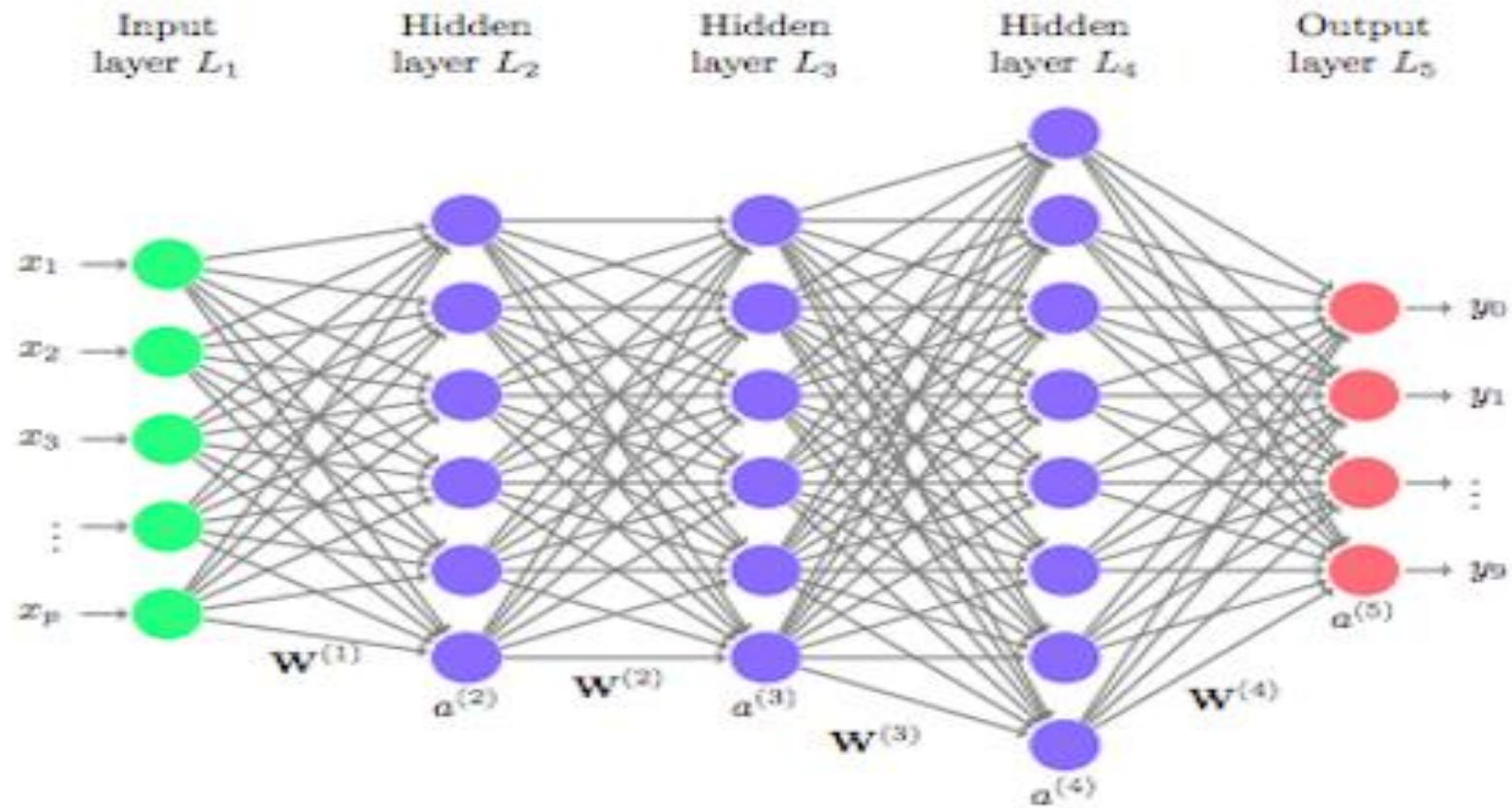
- In reality, it is not just a couple of neurons which would do the decision making.
- There is a massively parallel interconnected network of 10¹¹ neurons (100 billion) in our brain



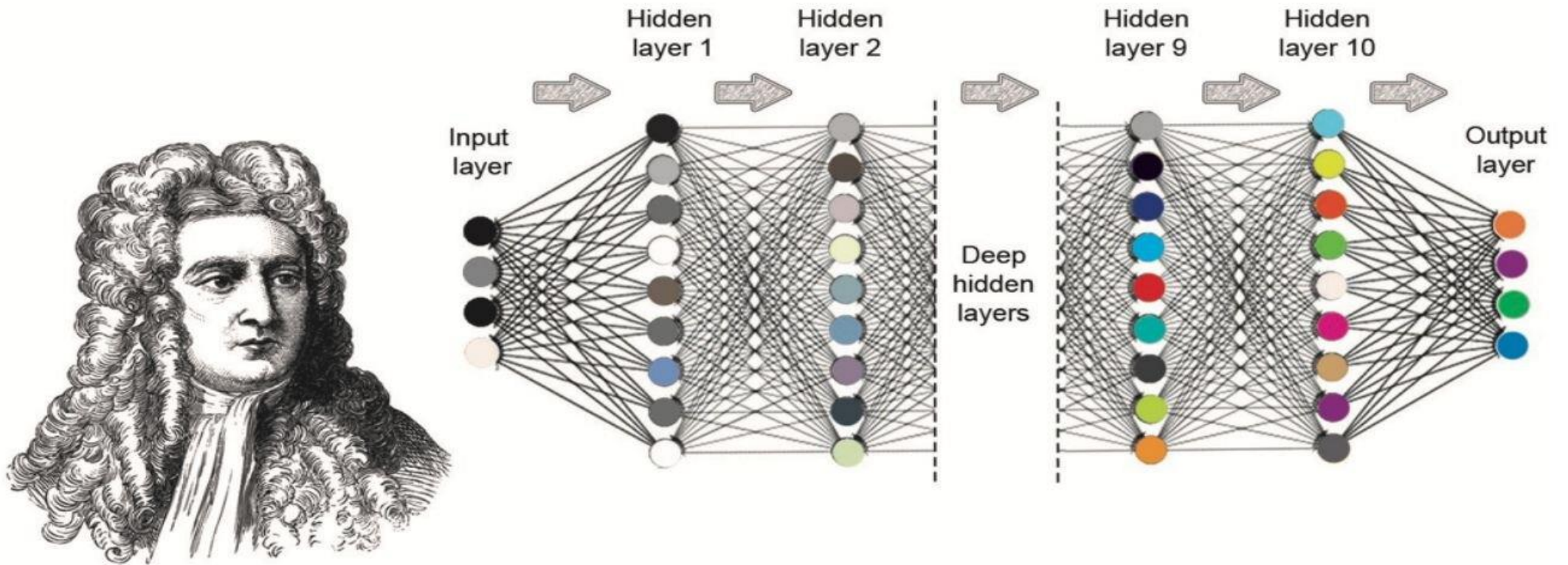
Multi Layer Perceptron (MLP)



Deep Neural Network(DNN)

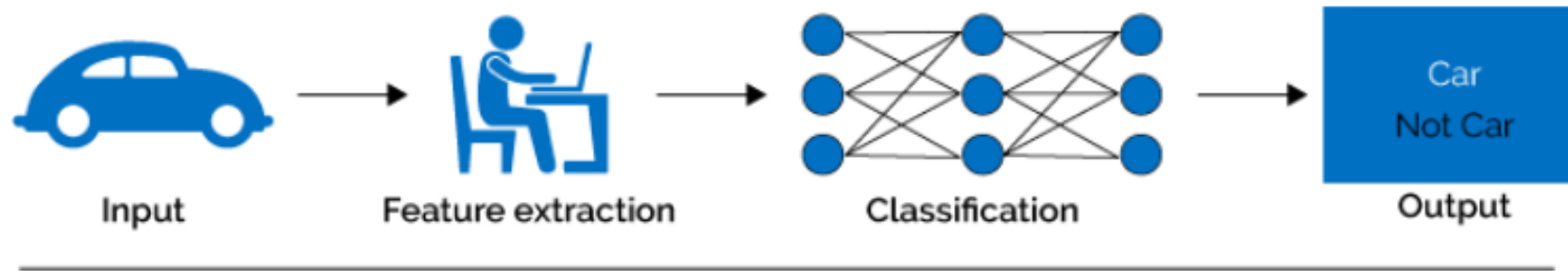


Deep Networks



ML V/S DL

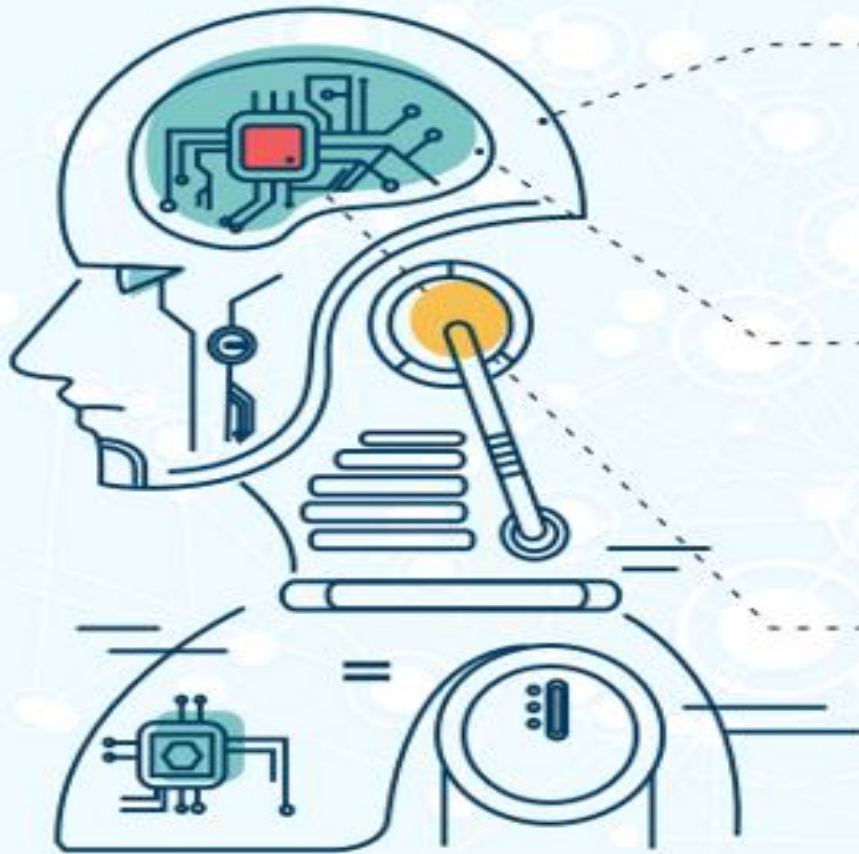
Machine Learning



Deep Learning



The differences between Artificial Intelligence, Machine Learning and Deep Learning



Artificial Intelligence (AI)

A field that studies how to create computer programs with **the ability to learn and reason like humans** in order to solve problems creatively.

Machine Learning (ML)

Application of AI dedicated to the creation of algorithms that allow systems **to learn without human intervention**, i.e. without the need for explicit programming.

Deep Learning (DL)

A subset of machine learning focused on **creating artificial neuronal networks**, in other words, systems that imitate the human brain, adapting and learning from large amounts of data.

Pros and cons of deep neural networks

Advantages

- Deep learning has solved problems in computer vision and audio much better than anything before. The same also holds for advanced **natural language** processing (e.g translation or speech-to-text). If you are facing any of those problems then this is the solution to go.
- The fact that it is still an active field, means that there is potential for **innovation**. Some startups have based their whole business model on creating deep learning models optimised for a single goal (e.g. detecting animals on images or translating from voice to text).
- Deep learning has been commoditised to some extent through services such as Cloud Vision, Cloud Translate or IBM's Watson. So, for some applications you might be able to roll something out quickly by using these services.

Pros and cons of deep neural networks

Drawbacks

- Requires large datasets. Not always available, especially for new business since either data might not exist (e.g. user data), or it might be too expensive.
- Requires lots of computing power, hence higher cost.
- Deep learning experts are a subset of the data science community. Hence, if there is shortage of data scientists, there is even larger shortage of deep learning experts.
- Even though there is some work being done in explaining the how deep neural nets learn, neural networks are still black boxes.
- Deep learning requires LOTS of tuning.

Three classes of DL

Supervised

- Training data
- Object & label

Unsupervised

- Labels not available

Hybrid

- Pre trained DNN